Open Set Cross-Domain Hyperspectral Image Classification Based on Critical Reflective Learning Network

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***Abstract*—Limited by the lack of supervisory information of unknown classes, the existing open set cross-domain hyperspectral image (HSI) classification methods often rely on threshold-based methods when identifying unknown classes and are difficult to adapt to the complex interclass variability between unknown and known classes in HSI. To solve this problem, this article proposes a critical-reflective learning network (CRLN) based on teacher–student network, which obtains the unknown classes probability from the prediction of the closed set classifier (teacher network) and provides supervisory information for the open set classifier (student network). Specifically, first, the teacher network is trained with source-domain data, and the unknown class probability predictions of the teacher network are obtained by quantifying the uncertainty of the network predictions. Second, the student network learns the complex boundary relationship between known and unknown classes based on the output of the teacher network to achieve accurate recognition of unknown classes. Furthermore, considering the credibility of the teacher network, a critical-reflective learning mechanism is proposed to allow the student network to reflect on the erroneous experiences of the teacher network, thus alleviating the performance damage caused by this potentially erroneous knowledge. Finally, the class alignment-separation module (CASM) is proposed, which uses contrastive learning to promote the separation of known and unknown classes in the feature space, so as to reduce the risk of negative transfer induced by the confusion of the two types of features during cross-domain distribution adaptation. Experiments on three datasets show that the proposed method achieves the state-of-the-art performance in the open set cross-domain HSI classification task.**

***Index Terms*—Cross-domain, domain adaptation, hyperspectral image (HSI), open set, teacher–student network.**

I. Introduction

In RECENT years, the hyperspectral image (HSI) classification methods based on deep learning have achieved outstanding performance in the fields of remote sensing interpretation [1], [2], [3], [4], environmental monitoring [5], [6], and land planning [7], [8], [9]. These methods usually assume that the training and test data follow the principle of independent and identical distribution. However, in real-world scenarios, due to differences in the environment, equipment, and other factors at the time of HSI acquisition, there often exists serious data distribution deviation [10] between the training scene (source domain) and the test scene (target domain). To this end, it is difficult to achieve satisfactory results with the existing methods.

Domain adaptation can alleviate the distribution differences between different HSIs through cross-domain distribution adaptation to ensure that the model trained on source-domain data can maintain good generalization performance in the target domain. Zhang et al. [11] proposed a graph information aggregation cross-domain few-shot learning framework, which effectively alleviated the problem of distribution differences between two domains by combining graph neural network information propagation and graph alignment strategy. However, it should be pointed out that the effectiveness of the existing domain adaptation methods is mostly based on the closed set assumption that the source domain and the target domain are of the same class [12], [13], [14], [15], and it is difficult to adapt to the open set scenarios, where there are unknown classes in the target domain.

As a result, the open set cross-domain HSI classification has emerged, which aims to extend the cross-domain scenario from closed set to open set, so that the model can identify the unknown classes of the target domain. The existing methods often rely on threshold-based solutions for unknown classes identification [16], [17], [18]. Specifically, the determination of the class of the samples is achieved by calculating the distance measure between samples, such as Euclidean distance or cosine distance, and comparing it with a predefined threshold. The determination of the threshold usually stems from the prototype calculation of the known class samples in the source-domain data or the estimate of the output probability. Although it is possible to distinguish known and unknown classes in feature space by setting appropriate thresholds, such methods often face the challenge of limited generalization performance in complex scenes with fuzzy boundaries between classes. Especially in the HSI classification, the complex interclass variability between known and unknown classes makes this challenge particularly severe. As shown in Fig. 1, in the Houston 2018 dataset, residential buildings and

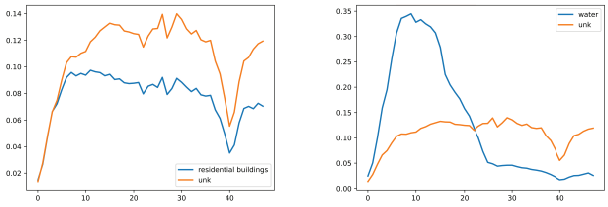
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(a) (b)

Fig. 1. Schematic of spectral similarity between known and unknown classes. The horizontal axis is the number of bands and the vertical axis is the spectral value. (a) Residential building (known class) and the unknown class (orange curve) show similar spectral curves. (b) Water (known class) and the unknown class (orange curve) show different spectral curves. In the band of 0–20, the spectral value of water (known class) increases first and then decreases, while the spectral value of the unknown class (orange curve) gradually increases.

unknown class exhibit similar spectral characteristics, which requires a smaller threshold for distinguishing between the two classes. In contrast, there is a large spectral difference between water and unknown class, and continuing to use a smaller threshold would lead to misclassification. At this time, the unknown class recognition mechanism based on fixed threshold is often difficult to adapt to the above two scenarios.

Based on the above analysis, how to break through the limitations of threshold-based schemes and develop an unknown class recognition mechanism that can learn the complex boundary relationship between unknown class and known classes is a key issue to be solved. The difficulty in solving this problem lies in that since the unknown classes are unknowable during training, the model lacks supervisory information about the unknown classes, which brings severe challenges to the identification of unknown classes [19], [20], [21]. To address the above problems, we use entropy as a measure of uncertainty. When the decision boundary is uncertain about the classification result of the target sample, higher entropy will be generated. In addition, for unknown classes, due to the lack of prior training examples, the classifier cannot accurately predict its class, which also leads to an increase in entropy. Based on this, the probability information of the unknown classes can be extracted from the prediction of the closed set classifier. Furthermore, the closed set classifier can be regarded as a teacher network, providing supervision signals for the open set classifier, and it can be used as a student network to learn the complex boundary relationship between known and unknown classes from the output of the teacher network.

To be detailed, a critical reflective learning network (CRLN) based on teacher–student network architecture is proposed in this article. First, the uncertainty difference between known and unknown classes in the output prediction of the pretrained teacher network is used to obtain the probability of the unknown classes, thereby providing the necessary supervision information for the student network. The student network can inherit this knowledge, quickly adapt to the target domain, and identify known/unknown class. Second, we introduce the critical reflective learning mechanism, which enables the student network to critically analyze the incorrect predictions of the teacher network. This process facilitates the student network learn from the experience of the teacher network while avoiding inheriting its mistakes, thereby reducing the risk of negative transfer. Finally, in order to reduce the risk of negative transfer caused by the confusion of unknown and known classes features during cross-domain distribution adaptation, a CASM is proposed. This module utilizes contrastive learning to promote the separation of known and unknown classes in the feature space. The contributions of this article can be summarized as follows.

1) CRLN is proposed, which introduces the teacher–student idea into the open set cross-domain HSI classification task, breaking through the limitation of open set classifiers relying on thresholds. Through the guidance of the teacher network, the student network can master the decision boundary of known and unknown classes.
2) An entropy determination module (EDM) is designed to obtain the confidence probability of unknown classes by quantifying the uncertainty of the teacher network's prediction output. Thereby, the teacher network has the ability to distinguish unknown classes and providing supervisory information for the student network to identify and distinguish known and unknown classes in the target domain.
3) A critical reflective learning mechanism is proposed to mitigate the problem of potential faulty knowledge in the teacher's network that would cause damage to the performance of the learning network through reflective learning from the teacher network's fallible samples. Thus, robust knowledge transfer between the teacher network and the student network can be achieved.
4) A CASM is designed, which aligns known classes and uses contrastive learning to promote the separation of unknown classes and known classes in the feature space. Alleviating the negative transfer phenomenon caused by feature confusion between known and unknown classes during cross-domain distribution adaptation.

II. Related Work

A. Closed Set Domain Adaptation

The closed set domain adaptation is a core direction in domain adaptation research [22], [23], [24], which aims to transfer knowledge from a domain with rich labels (source domain) to a domain with scarce labels (target domain). In this case, the two domains share the same set of classes. In other words, every sample in the target domain can theoretically be identified by a class in the source domain.

Many closed set domain adaptation methods focus on reducing the distribution difference between the source domain and the target domain. Fang et al. [25] proposed an unsupervised domain adaptation framework that combines masked self-distillation and class-separable adversarial training to extract class-related knowledge by performing pretext tasks on target domain samples. Li et al. [26] proposed a supervised contrastive learning-based unsupervised domain adaptation method, which improves the separability of the data itself within the domain by performing a supervised contrastive learning strategy in both the source and target

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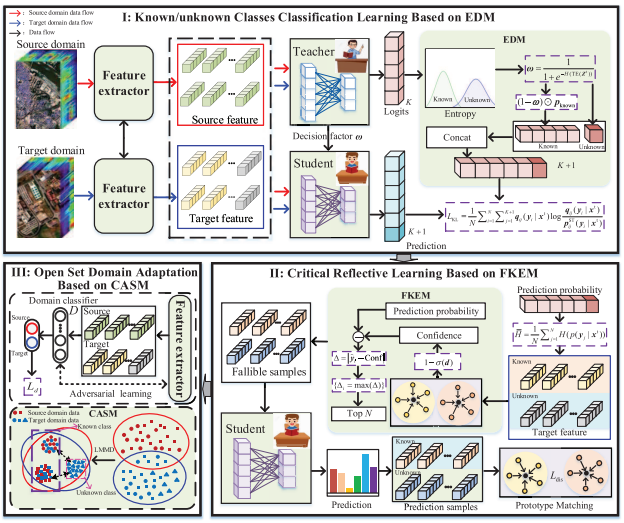


Fig. 2. Framework of CRLN. Stage I: the teacher network is trained on the source-domain data to generate predictions for known classes, and the uncertainty of the teacher network's predictions is quantified through the EDM to obtain the prediction probabilities of known and unknown classes. Then, the student network learns the classification of known and unknown classes from the teacher network's predictions based on L_{KL}. Stage II: FKEM is used to analyze and identify fallible samples of teacher network. Then, fallible samples are input into the student network to obtain prediction samples, which are used to calculate category prototypes and perform prototype matching. The above operations guide the student network to critically learn knowledge to improve its classification ability. Stage III: two domains are globally domain adaptation based on adversarial learning between feature extractor and domain classifier. The CASM aligns known classes in two domains based on subdomain correlation and pushes the unknown classes away from the known classes. The above operations ensure that the model can complete domain adaptation in an open set scenario.

domains and introduces a new one-class classification-based domain similarity loss to reduce interdomain differences. Ning et al. [27] proposed a class matching-based contrastive learning framework, which learns instance-level discriminative feature embedding by taking class information as a premise in the feature space, taking source samples as anchors, and searching for their positive and negative matching samples in the target domain. The essence of this framework is to extract features through the network and perform instance-level contrastive learning to reduce domain shift.

Closed set domain adaptation usually assumes that the class spaces of the source and target domains are completely consistent. This means that the model has been trained with all the classes in the target domain, which have corresponding samples in the source domain. However, in the real world, the target domain contains unknown classes that never appear in the source domain. Li et al. [28] proposed a universal domain adaptation framework, which achieved effective knowledge transfer between source and target domains without the need for target-domain labels through shared semantic pairing and domain similarity score. Zhang and Liu [16] designed a manifold embedding mapping to align the features of hyperspectral data in different scenes and selected target domain samples for pseudo-labeling in an iterative process, while rejecting samples of unknown classes.

B. Open Set Domain Adaptation

Compared with the classic closed set domain adaptation, the open set domain adaptation is a more realistic and more suitable problem setting for the real world [29], [30], [31]. In many practical applications, the target domain usually contains classes that do not appear in the source domain, which makes open set domain adaptation a more challenging and meaningful research direction. Since the open set problem involves identifying and adapting to unknown classes, it requires not only a strong feature representation capability but also a certain robustness and adaptability of the model to cope with the uncertainty and complexity brought by unknown classes.

Panareda Busto and Gall et al. [32] first proposed open set domain adaptation. They adopted a method called iterative assignment and transformation to assign labels to target data using the distance matrix between the target data and the source class center and the distribution aligned by the mapping matrix. Liu et al. [33] gradually separated the known and unknown classes in the target domain and simultaneously learned feature extractors and classifiers to adapt to open set domain adaptation tasks with different openness levels. Saito et al. [34] proposed an open set domain adaptation by backpropagation method based on adversarial learning, which forced the generator to either match the target sample with the source known classes or adapt it as an unknown class sample through adversarial training. Zhang et al. [35] aligned the source and target domains through adversarial learning and simultaneously used knowledge graphs and graph convolutional networks to generate classifiers for unknown class. Wang et al. [36] effectively transferred knowledge from labeled source graphs to unlabeled target graphs through a separate domain alignment strategy to achieve accurate classification of nodes of known classes and recognition of nodes of unknown classes. Wang et al. [37] combined contrastive self-supervised learning with consistency self-training and used contrastive self-supervised learning to mine the internal structural information of unlabeled target domain samples, obtain robust features, and then perform consistency self-training to reduce domain bias.

III. Method

A. Framework of the Proposed Method

In order to demonstrate the framework of CRLN, we take Pavia Center (PaviaC) used in the experiment in this article as the source domain and University of Pavia (PaviaU) as the target domain, and the framework of CRLN is shown in Fig. 2. Assume that there is a labeled source domain D_{s} and an unlabeled target domain D_{t}. Specifically, the source-domain sample set and source-domain label set are represented as $\boldsymbol{X}^{\mathrm{s}} = \{\boldsymbol{x}_1^{\mathrm{s}}, \boldsymbol{x}_2^{\mathrm{s}}, \ldots, \boldsymbol{x}_{n_{\mathrm{s}}}^{\mathrm{s}}\} \in \mathrm{R}^{n_{\mathrm{s}} \times d}$ and $\boldsymbol{Y}^{\mathrm{s}} = \{\boldsymbol{y}_1^{\mathrm{s}}, \boldsymbol{y}_2^{\mathrm{s}}, \ldots, \boldsymbol{y}_K^{\mathrm{s}}\}$, where n_{s} represents the number of source-domain samples, d represents spectral dimension, and K represents the number of classes. Similarly, the unlabeled target-domain sample data is represented as $\boldsymbol{X}^{\mathrm{t}} = \{\boldsymbol{x}_1^{\mathrm{t}}, \boldsymbol{x}_2^{\mathrm{t}}, \ldots, \boldsymbol{x}_{n_{\mathrm{t}}}^{\mathrm{t}}\} \in \mathrm{R}^{n_{\mathrm{t}} \times d}$, where n_{t} represents the number of target-domain samples. The algorithm flowchart is shown in Fig. 3, which includes two processes: training and testing.

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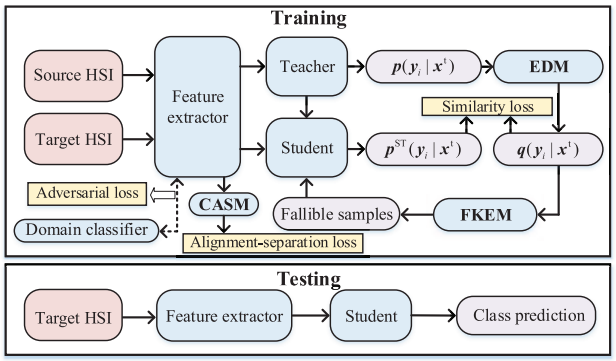


Fig. 3. Algorithm flowchart of CRLN.

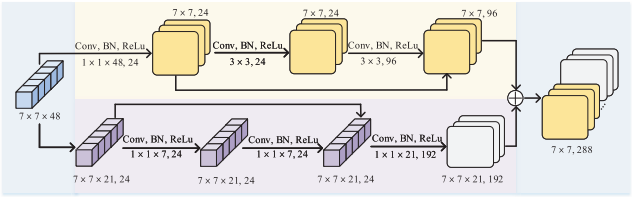


Fig. 4. Network structure of the feature extractor. Taking an input with a size of $7 \times 7 \times 48$ as an example, $\bigoplus$ represents a concatenation operator.

In the training process, first, after unifying the dimensions and partitioning the source- and target-domain data, the samples from the two domains are input into the feature extractor to obtain the spatial–spectral features. Second, the features are input into the teacher network and student network to obtain the predicted output $\boldsymbol{p}(\boldsymbol{y}_i|\boldsymbol{x}^t)$ and $\boldsymbol{p}^{ST}(\boldsymbol{y}_i|\boldsymbol{x}^t)$. Then, $\boldsymbol{p}(\boldsymbol{y}_i|\boldsymbol{x}^t)$ is inputted into EDM, and we can get the prediction probabilities of known and unknown classes $\boldsymbol{q}(\boldsymbol{y}_i|\boldsymbol{x}^t)$. Third, the student network learns knowledge by minimizing the similarity between $\boldsymbol{p}^{ST}(\boldsymbol{y}_i|\boldsymbol{x}^t)$ and $\boldsymbol{q}(\boldsymbol{y}_i|\boldsymbol{x}^t)$; then, we can get similarity loss. Since we aim to train a student network with excellent performance, the teacher network is not absolutely correct. If student network only follows the teacher network without distinguishing whether the knowledge is correct, the learning effect will be limited, so $\boldsymbol{q}(\boldsymbol{y}_i|\boldsymbol{x}^t)$ is inputted into fallible knowledge extraction module (FKEM) to explore the teacher network's fallible samples, and the student network can critically learn knowledge from the fallible samples. Finally, the global-domain adaptation is achieved through the adversarial process of domain classifier and feature extractor, so we can get the adversarial loss. The CASM aligns known classes in two domains based on subdomain correlation and pushes the unknown classes away from the known classes, and we get alignment-separation loss from CASM. Based on the above losses, the network performs gradient backpropagation and iterative optimization.

In the testing process, input target-domain HSI into the feature extractor. Based on the features, the trained student network is utilized to obtain class prediction.

B. Unknown Class Teaching Based on EDM

1) Feature Extraction: In this article, we use a spectral–spatial dual-branch feature extraction network as the feature extractor, and its structure is shown in Fig. 4. The upper branch extracts spatial features from the input data, and the lower branch is responsible for extracting spectral features from the network. Finally, the features obtained from the two branches are connected to obtain the final extracted features.

The feature extractor G is used to extract features from the data in the source domain and the target domain

$$\begin{cases} \boldsymbol{Z}^s = G\left(\boldsymbol{X}^s\right) \\ \boldsymbol{Z}^t = G\left(\boldsymbol{X}^t\right) \end{cases} \tag{1}$$

where $\boldsymbol{Z}^s$ and $\boldsymbol{Z}^t$ represent the embedded feature of the source domain and target domain.

2) Teacher–Student Network: After obtaining the embedded features of the two domains, the decision boundary is learned through the teacher network and the student network.

First, for the source-domain data, the labeled source-domain data are used to train the feature extractor and the teacher network, as shown in the following formula:

$$L_{TE} = \frac{1}{n_s} \sum_{\boldsymbol{x}^s, \boldsymbol{y}^s \in \boldsymbol{X}^s} L_{CE}\left(\text{Softmax}\left(\text{TE}\left(\boldsymbol{Z}^s\right), \boldsymbol{y}^s\right)\right) \tag{2}$$

where n_s is the number of samples in the source domain, TE represents the teacher network, and L_{CE} is the cross entropy loss, and its expression is

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} \boldsymbol{y}_{ic} \log\left(\boldsymbol{p}_{ic}\right) \tag{3}$$

where N is the number of samples in the batch, and $\boldsymbol{y}_{ic}$ and $\boldsymbol{p}_{ic}$ are, respectively, the true label of the ith sample belonging to the class c and the predicted probability.

Since the output dimension of the teacher network is the number of source-domain classes, if the target domain containing unknown classes is input into the teacher network, it will cause the unknown class samples to be incorrectly assigned to known classes. The general solution is to set a threshold to isolate unknown classes, but this method is prone to misclassification when the threshold is not selected properly. In this article, we use entropy as an indicator to identify unknown classes. Entropy is a measure of uncertainty. When the decision boundary of teacher network is uncertain about the classification result of the target sample, a higher entropy value will be generated. In addition, for unknown classes, due to the lack of prior training examples, the network cannot accurately predict its class, which also leads to an increase in entropy.

As a result, the teacher network assigns a higher entropy to the unknown classes than the known ones, thereby helping the model identify those unknown classes. In short, the entropy value reflects the uncertainty of the classification result and provides a solution for the identification of unknown classes. The output results of the teacher network are expanded using the EDM, so that it can identify unknown classes. EDM will be introduced in Section III-B3.

3) Entropy Determination Module: On the one hand, since there are no available labels in the target domain, entropy is used to assist the model in distinguishing known and unknown

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class samples, enhancing the robustness of the model when facing unknown class, and assisting in learning more accurate decision boundaries, especially between known and unknown classes. The entropy value is defined as the entropy of the probability distribution output by the teacher network after classifying the target-domain features extracted by the feature extractor G. The formula is as follows:

$$H\left(\mathrm{TE}\left(\boldsymbol{Z}^{\mathrm{t}}\right)\right) = -\sum_{i=1}^{K} \boldsymbol{p}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) \log \boldsymbol{p}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) \tag{4}$$

where $\boldsymbol{p}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$ is the predicted output by the teacher network that the sample output $\boldsymbol{x}^{\mathrm{t}}$ belongs to a class i.

Since teacher network TE assigns higher entropy to samples belonging to unknown classes in the target domain than to samples belonging to known classes, this provides a solution for unknown class recognition: when the network is certain about the classification result of a sample, the sample is considered to belong to a known class. When the network is uncertain about the classification result of a sample, the sample is considered to belong to an unknown class. This article designs a decision factor $\boldsymbol{\omega}$ to quantify this certainty and uses it to estimate the probability that the sample belongs to an unknown class. In order to facilitate the probability estimation of unknown classes, we use the standard sigmoid function $\sigma(\boldsymbol{u}) = (1/1 + e^{-\boldsymbol{u}})$ to limit $\boldsymbol{\omega}$ to the range of (0, 1), so we define the decision factor $\boldsymbol{\omega}$ as

$$\boldsymbol{\omega} = \frac{1}{1 + e^{-H(\mathrm{TE}(\boldsymbol{Z}^{\mathrm{t}}))}}. \tag{5}$$

Analyzing the above formula, $\boldsymbol{u}$ in the sigmoid function corresponds to $H(\mathrm{TE}(\boldsymbol{Z}^{\mathrm{t}}))$. Since the value of $H(\mathrm{TE}(\boldsymbol{Z}^{\mathrm{t}}))$ is nonnegative, so $\boldsymbol{\omega}$ is a value between 0.5 and 1, which approximately represents the probability that the samples $\boldsymbol{x}^{\mathrm{t}}$ belong to an unknown class. When the entropy value is low (that is, the network is certain about the classification result), $\boldsymbol{\omega}$ is close to 0.5, indicating that the samples belong to a known class. When the entropy value is high (that is, the network is uncertain about the classification result), $\boldsymbol{\omega}$ is close to 1, indicating that the samples belong to an unknown class.

For the student network, first, the source classification loss is constructed on $\boldsymbol{X}^{\mathrm{s}}$

$$L_{\mathrm{cls1}} = \frac{1}{n_{\mathrm{s}}} \sum_{\boldsymbol{x}^{\mathrm{s}}, \boldsymbol{y}^{\mathrm{s}} \in \boldsymbol{X}^{\mathrm{s}}} L_{\mathrm{CE}}\left(\mathrm{Softmax}\left(\mathrm{ST}\left(\boldsymbol{Z}^{\mathrm{s}}\right), \boldsymbol{y}^{\mathrm{s}}\right)\right) \tag{6}$$

where ST represents the student network. Second, for the target domain, student network is required to learn the classification boundaries of unknown classes

$$L_{\mathrm{cls2}} = \frac{1}{n_{\mathrm{t}}} \sum_{\boldsymbol{x}^{\mathrm{t}} \in \boldsymbol{X}^{\mathrm{t}}} \boldsymbol{\omega} L_{\mathrm{CE}}\left(\mathrm{Softmax}\left(\mathrm{ST}\left(\boldsymbol{Z}^{\mathrm{t}}\right), \boldsymbol{y}^{\mathrm{unk}}\right)\right) \tag{7}$$

where $\boldsymbol{y}^{\mathrm{unk}}$ is the one-hot encoding of the unknown classes. The total classification loss of student network can be expressed as

$$L_{\mathrm{cls}} = L_{\mathrm{cls1}} + L_{\mathrm{cls2}}. \tag{8}$$

On the other hand, since the output dimension of the teacher network is a known number of class K, student network cannot directly learn the teacher network's output knowledge. Therefore, we first expand its output dimension to $K + 1$

$$\boldsymbol{p}_{\mathrm{known}}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) = \boldsymbol{p}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) \odot (1 - \boldsymbol{\omega}), \quad i = 1, 2, \ldots, K \tag{9}$$

where $\odot$ represents the elementwise product.

Third, $\boldsymbol{\omega}$ is used as an additional dimension to represent the probability of the unknown classes, thereby expanding the output dimension of the teacher network from K to $K+1$ and constructing a new prediction probabilities $\boldsymbol{q}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$

$$\boldsymbol{q}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) = \boldsymbol{p}_{\mathrm{known}}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}} : \boldsymbol{\omega}\right), \quad i = 1, 2, \ldots, K. \tag{10}$$

Finally, the student network learns by minimizing the KL divergence between $\boldsymbol{p}_{ij}^{\mathrm{ST}}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$ and $\boldsymbol{q}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$ of the teacher network

$$\begin{aligned} &L_{\mathrm{KL}}\left(\boldsymbol{q}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right)|\boldsymbol{p}^{\mathrm{ST}}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right)\right) \\ &= \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{K+1} \boldsymbol{q}_{ij}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right) \log \frac{\boldsymbol{q}_{ij}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right)}{\boldsymbol{p}_{ij}^{\mathrm{ST}}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right)} \end{aligned} \tag{11}$$

where N is the batch size, $K+1$ is the total number of classes, $\boldsymbol{p}_{ij}^{\mathrm{ST}}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$ is the probability that the ith sample belongs to the jth class, and $\boldsymbol{q}_{ij}(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}})$ is the probability that the ith sample belongs to the jth class.

In this way, the student network is able to learn not only the knowledge of the teacher network about the known classes but also how to deal with the unknown classes, thus improving the classification ability of the model on the target domain.

In summary, the total loss at this stage can be expressed as

$$L_{\mathrm{c}} = L_{\mathrm{TE}} + L_{\mathrm{cls}} + \boldsymbol{\alpha} L_{\mathrm{KL}} \tag{12}$$

where $\boldsymbol{\alpha}$ represents the loss weight coefficient.

C. Critical Reflective Learning Based on FKEM

Since the teacher network's knowledge is not absolutely correct, if the student network indiscriminately accepts the guidance of the teacher network without considering the possible knowledge errors or limitations of the teacher network, the student network's learning results will inevitably be negatively affected. Specifically, the teacher network will make wrong predictions when the boundaries of some classes are fuzzy, which lead to the student network's learning bias on these classes, thereby affecting its classification accuracy for new samples.

Therefore, critical learning is proposed to identify the mistakes that the teacher network make, so as to extract more accurate knowledge. Specifically, by analyzing the prediction results of the teacher network, we can identify the samples that are most likely to be wrong in the teacher network's prediction, which represent the knowledge blind spots of the teacher network. For example, the teacher network may misclassify similar classes in some complex scenarios, causing the student network to learn incorrect features on these classes.

Next, the student network is guided to deeply learn these fallible samples. In this way, the student network can not only correct the mistakes of the teacher network but also further improve its recognition and generalization capabilities based on the teacher network, especially when dealing with samples

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that the teacher network has difficulty in correctly classifying. This method enables the student network to have a certain critical thinking in the learning process, thereby improving its ability to adapt to complex and unknown classes.

1) Fallible Knowledge Extraction Module: We use FKEM to identify the samples that the teacher network is most likely to make mistakes. Specifically, first, the samples are divided into known and unknown classes based on the average entropy value

$$\begin{cases} \bar{H} = \dfrac{1}{N}\displaystyle\sum_{j=1}^{N} H\left(\boldsymbol{p}\left(\boldsymbol{y}_i|\boldsymbol{x}^{\mathrm{t}}\right)\right), \boldsymbol{x}^{\mathrm{t}} \in \boldsymbol{X}^{\mathrm{t}} \\ \boldsymbol{X}^{\mathrm{t}} = \boldsymbol{X}_{\mathrm{k}} \cup \boldsymbol{X}_{\mathrm{unk}} \\ \boldsymbol{X}_{\mathrm{k}} = \left\{\boldsymbol{x}_{\mathrm{k}} > \bar{H}\right\} \\ \boldsymbol{X}_{\mathrm{unk}} = \left\{\boldsymbol{x}_{\mathrm{unk}}|\boldsymbol{x}^{\mathrm{t}} <= \bar{H}\right\} \end{cases} \tag{13}$$

where $\boldsymbol{X}_{\mathrm{k}}$ is the set of known samples for the target domain, $\boldsymbol{X}_{\mathrm{unk}}$ is the set of unknown samples for the target domain.

Second, for each example $\boldsymbol{x}^{\mathrm{t}}$, we compute the sum of the Euclidean distances $\boldsymbol{d}_{\mathrm{k}}$ and $\boldsymbol{d}_{\mathrm{unk}}$ to the known class center $\boldsymbol{c}_{\mathrm{k}}$ and the unknown class center $\boldsymbol{c}_{\mathrm{unk}}$

$$\begin{cases} \boldsymbol{d}_{\mathrm{k}} = \left\|\boldsymbol{z}^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{k}}\right\|_2 \\ \boldsymbol{d}_{\mathrm{unk}} = \left\|\boldsymbol{z}^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{unk}}\right\|_2 \end{cases} \tag{14}$$

where $\|\boldsymbol{z}^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{k}}\|_2 = \left(\sum_{i=1}^{D}\left(\boldsymbol{z}_i^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{k},i}\right)^2\right)^{1/2}$, $\|\boldsymbol{z}^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{unk}}\|_2 = \left(\sum_{i=1}^{D}\left(\boldsymbol{z}_i^{\mathrm{t}} - \boldsymbol{c}_{\mathrm{unk},i}\right)^2\right)^{1/2}$, D is the dimension of the feature vector, $\boldsymbol{c}_{\mathrm{k},i}$, and $\boldsymbol{c}_{\mathrm{unk},i}$ is the value of the class center in the first dimension.

Third, concatenating $\boldsymbol{d}_{\mathrm{k}}$ and $\boldsymbol{d}_{\mathrm{unk}}$ to get the distance set $\boldsymbol{d}$ of the known and unknown classes samples predicted in a batch to the center of their respective classes, and convert $\boldsymbol{d}$ to the confidence that the sample belongs to the class through the sigmoid function. The confidence reflects the closeness of the sample to the class center, where a smaller corresponds to a higher probability, meaning that the sample is more likely to belong to the class. The calculation formula is

$$\begin{cases} \boldsymbol{d} = \mathrm{CONCAT}\left(\boldsymbol{d}_{\mathrm{k}}, \boldsymbol{d}_{\mathrm{unk}}\right) \\ \mathrm{Conf} = 1 - \boldsymbol{\sigma}\left(\boldsymbol{d}\right) \end{cases} \tag{15}$$

where CONCAT$(\cdot)$ represents the concatenation operation and $\boldsymbol{\sigma}(\boldsymbol{d}) = (1/1 + e^{-\boldsymbol{d}})$.

Fourth, the difference between these confidences and the predicted probabilities $\widehat{\boldsymbol{y}}_{\mathrm{t}}$ of the teacher network is calculated to identify the samples with the largest classification uncertainty, that is, the least confident

$$\Delta = \left|\widehat{\boldsymbol{y}}_{\mathrm{t}} - \mathrm{Conf}\right|. \tag{16}$$

Finally, the top N samples with the largest differences are selected as fallible samples. These samples are the focus of further learning of the student network

$$\boldsymbol{X}_{\mathrm{fault}} = \left\{\boldsymbol{x}_i|\Delta_i = \max\left(\Delta\right), i \in \{1, 2, \ldots, N\}\right\} \tag{17}$$

where $\boldsymbol{X}_{\mathrm{fault}}$ represents the set of fallible samples.

2) Critical Reflective Learning: This section proposes a refined student network learning method that focuses on learning from the fallible samples identified by the teacher network. These fallible samples are key samples that the teacher network could not accurately classify, and these samples are first fed into the student network to obtain the predicted probability output $\boldsymbol{O}$

$$\boldsymbol{O} = \mathrm{Softmax}\left(\mathrm{ST}\left(G\left(\boldsymbol{X}_{\mathrm{fault}}\right)\right)\right). \tag{18}$$

The predicted class index $\boldsymbol{h}_{\mathrm{indices}}$ for each sample can then be determined as

$$\boldsymbol{h}_{\mathrm{indices}} = \mathrm{argmax}\left(\boldsymbol{O}\right). \tag{19}$$

Based on the predicted class index $\boldsymbol{h}_{\mathrm{indices}}$ of the student network, the fallible samples were divided into known class samples $\boldsymbol{X}_{\mathrm{k}}$ and unknown class samples $\boldsymbol{X}_{\mathrm{unk}}$. Next, we compute the distance between these samples and the center of their class. Finally, we compute the sum of the distances between all samples and the center of their class and use this as the distance-based loss function L_{dis}

$$L_{\mathrm{dis}} = \sum\left(\left\|\boldsymbol{z}_{\mathrm{k}}^{\mathrm{ST}} - \boldsymbol{c}_{\mathrm{k}}^{\mathrm{ST}}\right\|_2 + \left\|\boldsymbol{z}_{\mathrm{unk}}^{\mathrm{ST}} - \boldsymbol{c}_{\mathrm{unk}}^{\mathrm{ST}}\right\|_2\right) \tag{20}$$

where $\boldsymbol{z}_{\mathrm{k}}^{\mathrm{ST}}$ and $\boldsymbol{z}_{\mathrm{unk}}^{\mathrm{ST}}$ are the known and unknown class samples output by the student network, respectively, and $\boldsymbol{c}_{\mathrm{k}}^{\mathrm{ST}}$ and $\boldsymbol{c}_{\mathrm{k}}^{\mathrm{ST}}$ represent the class centers of the known and unknown classes, respectively.

Through this strategy, even in the face of the knowledge limitations of the teacher network, the student network is able to focus on learning samples that the teacher network failed to classify accurately, thereby achieving more accurate classification and more robust performance in open set recognition tasks. By optimizing L_{dis}, the student network is able to learn how to more accurately map the feature representation of the sample to the center position of its predicted class, thereby enhancing the model's ability to identify and classify those challenging fallible samples.

D. Open Set Domain Adaptation Based on CASM

1) Global Domain Adaptation: We use adversarial learning to help align source and target domains globally. To be more specific, the feature extractor G tries to learn a domain-invariant feature representation, making it difficult for the domain classifier D to distinguish the features of the source and target domains, while the domain classifier tries to maximize its ability to distinguish the source domain and the target domain

$$\min_G \max_D \left\{\mathbb{E}_{\boldsymbol{p}(\boldsymbol{x},\boldsymbol{y})}\left[L_{\mathrm{CE}}\left(\mathrm{ST}\left(\boldsymbol{z}^{\mathrm{s}}\right), \boldsymbol{y}^{\mathrm{s}}\right)\right] + L_d\right\}. \tag{21}$$

The domain adversarial loss can be defined as

$$L_d = \mathbb{E}_{\boldsymbol{x}^{\mathrm{s}}\sim\boldsymbol{p}_{\mathrm{s}}}\left[\log D\left(\boldsymbol{z}^{\mathrm{s}}\right)\right] + \mathbb{E}_{\boldsymbol{x}^{\mathrm{t}}\sim\boldsymbol{p}_{\mathrm{t}}}\left[\log\left(1 - D\left(\boldsymbol{z}^{\mathrm{t}}\right)\right)\right] \tag{22}$$

where $\boldsymbol{p}_{\mathrm{s}}$ and $\boldsymbol{p}_{\mathrm{t}}$ are the data distributions of the source and target domains, respectively. $\boldsymbol{z}^{\mathrm{s}}$ and $\boldsymbol{z}^{\mathrm{t}}$ are the features extracted by the feature extractor G for source-domain sample $\boldsymbol{x}^{\mathrm{s}}$ and target-domain sample $\boldsymbol{x}^{\mathrm{t}}$.

It is worth noting that the adversarial process between the domain classifier and the feature extractor is carried out alternately.

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Stage I: (Update the Domain Classifier D): Fix the parameters of the feature extractor G and update the parameters of the domain classifier D to maximize the domain adversarial loss L_d

$$\max_{\theta_d} L_d = \mathbb{E}_{\boldsymbol{x}^{\mathrm{s}}\sim \boldsymbol{p}_{\mathrm{s}}}\left[\log D\left(\boldsymbol{z}^{\mathrm{s}}\right)\right] + \mathbb{E}_{\boldsymbol{x}^{\mathrm{t}}\sim \boldsymbol{p}_{\mathrm{t}}}\left[\log\left(1 - D\left(\boldsymbol{z}^{\mathrm{t}}\right)\right)\right] \quad (23)$$

where θ_d is the network parameter of the domain classifier.

Stage II: (Update Feature Extractor G): Fix the parameters of domain classifier D and update the parameters of feature extractor G to minimize domain adversarial loss L_d

$$\min_{\theta_{\mathrm{g}}} L_d = \mathbb{E}_{\boldsymbol{x}^{\mathrm{s}}\sim \boldsymbol{p}_{\mathrm{s}}}\left[\log D\left(\boldsymbol{z}^{\mathrm{s}}\right)\right] + \mathbb{E}_{\boldsymbol{x}^{\mathrm{t}}\sim \boldsymbol{p}_{\mathrm{t}}}\left[\log\left(1 - D\left(\boldsymbol{z}^{\mathrm{t}}\right)\right)\right] \quad (24)$$

where θ_{g} is the network parameter of the feature extractor. It is worth noting that in stage II of adversarial training, the optimization process of the feature extractor is not updated separately but is updated jointly with other losses.

2) Class Alignment-Separation Module:

a) Alignment of Known Classes: If we only focus on global adaptation and ignore the correlation between subdomains belonging to the same class across domains, some irrelevant data are too close, thus affecting the accuracy of classification. In order to overcome the limitations of global domain adaptation, subdomain adaptation methods are also needed. According to the class of the samples, the source domain and the target domain are divided into subdomains. The prediction results of the target-domain samples are output by the student network, and the known class samples are screened out and aligned with the known class of the source domain by LMMD

$$L_{\mathrm{LMMD}}\left(\boldsymbol{X}^{\mathrm{s}}, \boldsymbol{X}^{\mathrm{t}}, \boldsymbol{Y}^{\mathrm{s}}, \boldsymbol{Y}^{\mathrm{t}}\right) = \frac{1}{K}\sum_{k=1}^{K}\left\|\sum_{\boldsymbol{x}_i^{\mathrm{s}}\in\mathcal{D}_{\mathrm{s}}} \boldsymbol{z}_i^{\mathrm{s}} - \sum_{\boldsymbol{x}_j^{\mathrm{t}}\in\mathcal{D}_{\mathrm{t}}} \boldsymbol{z}_i^{\mathrm{t}}\right\|_{\mathcal{H}}^2 \quad (25)$$

where $k \in \{1, 2, \ldots, K\}$ is the class index and $\boldsymbol{z}_i$ is the two-domain features obtained by the feature extractor.

II. *Unknown/Known Separation*: In order to better identify unknown classes, it is necessary to separate known classes from unknown classes. A contrastive loss function is introduced to optimize the feature representation of the student network. This loss function pays special attention to the same classes in the source and target domains, and by calculating the similarity between sample features, it encourages the model to effectively distinguish unknown classes from known classes.

First, a Gaussian kernel function is defined to calculate the similarity matrix between samples in the source-domain features $\boldsymbol{Z}^s$ and the target-domain features $\boldsymbol{Z}^t$

$$K\left(\boldsymbol{x}_i, \boldsymbol{x}_j\right) = \exp\left(-\frac{\|\boldsymbol{z}_i - \boldsymbol{z}_j\|^2}{2\boldsymbol{\mu}^2}\right) \quad (26)$$

where $\boldsymbol{\mu}$ is the bandwidth parameter of the Gaussian kernel.

Second, the contrast loss function can be expressed as

$$L_{\mathrm{con}} = \sum_{i=1}^{N}\sum_{p\in\boldsymbol{p}(i)} -\log\frac{\exp\left(\frac{K(\boldsymbol{x}_i, \boldsymbol{x}_p)}{\tau}\right)}{\sum_{a\in\boldsymbol{A}(i)}\exp\left(\frac{K(\boldsymbol{x}_i, \boldsymbol{x}_{\mathrm{a}})}{\tau}\right)} \quad (27)$$

where $K(\boldsymbol{x}_i, \boldsymbol{x}_{\mathrm{p}})$ denotes the Gaussian kernel similarity between a sample $\boldsymbol{x}_i$ and its positive sample $\boldsymbol{x}_{\mathrm{p}}$, and $K(\boldsymbol{x}_i, \boldsymbol{x}_{\mathrm{a}})$ denotes the Gaussian kernel similarity between a sample $\boldsymbol{x}_i$ and negative sample $\boldsymbol{x}_{\mathrm{a}}$. τ is the temperature parameter. $\boldsymbol{p}(i)$ is the set of sample indexes of the same class as the sample $\boldsymbol{x}_i$, and $\boldsymbol{A}(i)$ is the set of all sample indexes except the sample $\boldsymbol{x}_i$.

This loss function encourages the model to bring samples belonging to the same class closer together in the feature space, while pushing samples of different classes farther away. In this way, the student network can learn clearer class boundaries in the feature space, thereby improving the model's ability to recognize known and unknown samples. For simplicity, we call L_{LMMD} and L_{con} as alignment-separation loss.

In summary, the loss of the domain adaptation part can be expressed as

$$L_{\mathrm{D}} = L_d + L_{\mathrm{LMMD}} + L_{\mathrm{con}}. \quad (28)$$

E. Total Loss

The total loss of the network is as follows:

$$L_{\mathrm{total}} = L_{\mathrm{c}} + \boldsymbol{\beta}_1 L_{\mathrm{dis}} + \boldsymbol{\beta}_2 L_{\mathrm{D}} \quad (29)$$

where $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are the weight coefficients, which are used to balance the weights of classification tasks, reflective learning, and domain adaptation tasks.

IV. Experiment

A. Dataset

Houston dataset: We used high-resolution hyperspectral datasets collected on the University of Houston campus and its surrounding areas: Houston 2013 and Houston 2018. The Houston 2013 dataset contains 144 spectral bands with a wavelength range of 380~1050 nm and a spatial resolution of 2.5 m. The Houston 2013 dataset contains 144 spectral bands ranging from 380 to 1050 nm with a spatial resolution of 2.5 m. The Houston 2018 dataset consists of 48 spectral bands covering the same wavelength range with a spatial resolution of 1 m. According to the experimental settings of the open set, we select a portion of the classes from each dataset as unknown classes. During training, only the 48 bands from the Houston 2013 dataset are used to ensure consistency between the source and target domains. The class names and their numbers in the dataset are shown in Table I. It is worth mentioning that the source domain is Houston 2013 and the target domain is Houston 2018.

Botswana dataset (BOT5 and BOT6): Captured in May and June of 2001, respectively, using the Hyperion sensor in the same area of Okavango, Botswana, South Africa. Both images of HSI contain 1476 × 256 pixels with 242 bands. The two images have nine identical ground feature classes, including: Firescar, Island interior, Riparian, Primary floodplain, Woodlands, Water, Savanna, Short mopane, and Exposed soils. We set Savanna, Short mopane, and Exposed soils as unknown classes. The Botswana dataset description is shown in Table I. It is worth mentioning that the source domain is BOT5 and the target domain is BOT6.

Pavia Dataset: The dataset consists of two subdatasets: the PaviaU dataset and the PaviaC dataset. The PaviaU dataset

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TABLE I
CLASSES AND THEIR NUMBER OF THREE DATASETS

		HS13	HS18			BOT5	BOT6			PaviaC	PaviaU
Class	Name	Number	Number	Class	Name	Number	Number	Class	Name	Number	Number
H1	Grass healthy	345	1353	B1	Water	297	361	P1	Tree	7598	3064
H2	Grass stressed	365	4888	B2	Primary floodplain	437	308	P2	Asphalt	9248	6631
H3	Trees	365	2766	B3	Riparian	448	303	P3	Brick	2685	3682
H4	Water	285	22	B4	Firescar	354	335	P4	Bitumen	7282	1330
H5	Residential buildings	319	5347	B5	Island interior	337	370	P5	Shadow	2683	947
				B6	Woodlands	357	324				
Unk	Road	0	6365	Unk	Savanna Short mopane Exposed soils	0	870	Unk	Meadow Bare soil	0	23678
	Total	1679	20741		Total	2230	2871		Total	29496	39332

contains pixels and 103 spectral bands, while the PaviaC dataset contains pixels and 102 spectral bands. To ensure that the two datasets have the same number of spectral bands, we delete the last spectral band from the Pavia dataset according to the experimental settings of [26]. The classes in the dataset and their numbers are shown in Table I. It is worth mentioning that the source domain is PaviaC and the target domain is PaviaU.

B. Experimental Setting

All experiments were conducted on a computer equipped with a 5.70-GHz AMD Ryzen 9 9950X CPU, 64-GB RAM, and RTX 4080 SUPER GPU, and the experimental environment was PyTorch. The comparative methods in this study include closed set domain adaptation methods in the field of computer vision: MSDA [25]; closed set domain adaptation methods in HSI: MLUDA [38], SCLUDA [26]; open set domain adaptation methods in the field of computer vision: SSOUDA [37], UADAL [39], and OSLPP [40].

To ensure fair comparison, the labels of the target domain are invisible in all experiments. For closed set domain adaptation methods, we use a threshold to distinguish known classes from unknown classes. When training CRLN, we use SGD as the optimizer with a learning rate of 0.001 and a patch size of 7. To eliminate the impact of randomness on the experiment, all results are the average of ten runs. Four evaluation indicators are selected, including OS, OS*, HOS, and kappa coefficient, where OS is the accuracy calculated on all the classes, OS* is the accuracy calculated on the target-domain known classes, UNK is the classification accuracy of the unknown class, and HOS is the harmonic mean of OS* and UNK.

C. Parameter Analysis and Setting

In this section, we analyzed the impact of different parameter settings on the performance of CRLN on three datasets, including α, β_1, β_2, and τ. We utilize HOS, which is a harmonic mean of OS* and UNK [41]. OS* is a classwise averaged accuracy over known classes, and UNK is an accuracy only for the unknown class. HOS is suitable for open set domain adaptation because it is higher when performing well in both known and unknown classifications. As shown in Fig. 5, the parameter settings are analyzed on the three datasets.

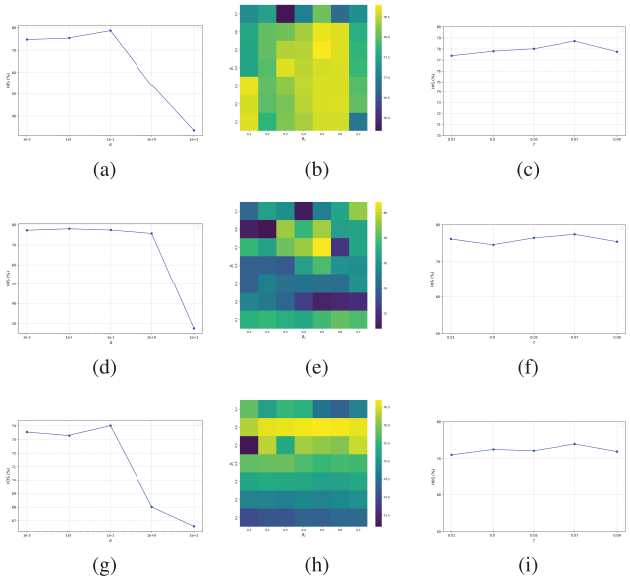


Fig. 5. Variation of HOS with different parameters. (a) Variation of HOS with different α on Houston dataset. (b) Variation of HOS with different β_1 and β_2 on Houston dataset. (c) Variation of HOS with different τ on Houston dataset. (d) Variation of HOS with different α on BOT dataset. (e) Variation of HOS with different β_1 and β_2 on BOT dataset. (f) Variation of HOS with different τ on BOT dataset. (g) Variation of HOS with different α on Pavia dataset. (h) Variation of HOS with different β_1 and β_2 on Pavia dataset. (i) Variation of HOS with different τ on Pavia dataset.

First, we analyzed the parameter α. α is the weight coefficient of the KL divergence loss of the teacher–student output, which is used to measure the difference between the output distributions of the teacher network and the student network. As can be seen from Fig. 5: as α increases, HOS shows a trend of first increasing and then decreasing. This is because: within a smaller range of α values, appropriately increasing α can enhance the student network's ability to learn the teacher network's knowledge. However, when α exceeds $1e^{-1}$, HOS begins to drop sharply. This is because an excessively large α value causes the student network to focus too much on fitting the output of the teacher network during training, while ignoring the independent learning of the data itself.

In the BOT dataset, HOS reaches the highest value when $\alpha = 1e^{-2}$ in Fig. 5(d). In the Houston dataset and the Pavia dataset, HOS is maximum when $\alpha = 1e^{-1}$ in Fig. 5(a) and (g). This is because the BOT dataset consists of HSIs collected in May and June of the same year using the same sensor

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in the same region. The difference between the two domains is relatively small. Therefore, the student network does not need to rely heavily on the experience of the teacher network, resulting in a smaller optimal value of α. In contrast, the HSIs of Houston dataset were collected by researchers in 2013 and 2018 using different sensors. Although they belong to the same region, the long collection time interval introduces significant differences between the domains, and the Pavia dataset was collected in two different regions using the same sensor, which also leads to substantial differences between the two domains. Compared with the BOT dataset, the student network needs to leverage the knowledge from the teacher network to learn interclass knowledge more extensively, resulting in a larger optimal value of α.

Then, we studied the effects of β_1 and β_2 on HOS, where β_1 and β_2 are the weight coefficients of the distance-based loss and the domain adaptation loss, respectively. The colors in Fig. 5(b), (e), and (h) represent the values of HOS: the color changes from dark purple to yellow, and HOS gradually increases. β_1 that is too large will cause the network to pay too much attention to the fault samples, thus affecting the classification performance of method. β_1 that is too small will cause the network to be unable to fully utilize the fault experience of the teacher network. β_2 that is too large may cause the network to focus too much on domain adaptation and ignore the classification task, while β_2 that is too small will cause the network to be unable to effectively adapt to the differences between domains, affecting the network's recognition of unknown classes.

For the Houston dataset, the BOT dataset, and the Pavia dataset, it can be seen from Fig. 5(b), (e), and (h) that HOS reaches its maximum value at $\beta_2 = 0.5$. For the Houston dataset and the BOT dataset, HOS reaches its maximum value at $\beta_1 = 0.5$. For the Pavia dataset, it can be seen from Fig. 5(h) that HOS reaches its maximum value at $\beta_1 = 0.6$. This is because, on the one hand, the Pavia dataset was collected in two different regions, and the domain difference of the two domain is large. On the other hand, since there are many samples of unknown classes in the target domain, accounting for about 60% of the total number of samples, which is much larger than the proportion of unknown classes in the Houston and BOT datasets, the model will generate more fault samples during training, which causes the student network to rely more on the teacher's knowledge and fault experience to identify known and unknown classes. Therefore, while during domain adaptation, the model needs to pay more attention to utilize the teacher network's fault experience to help the student network complete the classification task.

Finally, we studied the effect of temperature τ on HOS. As can be seen from Fig. 5(c), (f), and (i): for the three datasets, HOS generally shows a trend of increasing first and then decreasing. As τ increases, HOS reaches a peak at $\tau = 0.07$ and then begins to decline. The reason for the above trend is that when τ is low, the similarity of positive sample pairs will be amplified, making it easier for the network to distinguish between positive and negative sample pairs, thereby improving its HOS. When τ is high, the similarity of positive sample pairs decreases, while the similarity of negative sample pairs will be amplified, which will increase the difficulty of the network to distinguish between positive and negative sample pairs, and HOS will also decrease. When $\tau = 0.07$, HOS reaches its maximum value.

TABLE II
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (HOUSTON DATASET)

Class	SCLUDA	MLUDA	MSDA	OSLPP	SSOUDA	UADAL	CRLN
H1	88.71±1.33	84.17±3.69	88.17±4.89	93.86±7.79	**97.02±0.76**	91.07±7.47	88.52±3.52
H2	36.56±1.61	39.54±5.21	44.24±3.09	37.15±4.97	38.98±1.61	52.56±4.50	**62.94±2.00**
H3	43.15±3.37	37.65±6.09	**66.88±2.41**	62.69±1.52	64.46±0.97	65.61±3.10	66.46±2.78
H4	45.45±0.28	90.91±7.42	81.67±9.32	**100±0**	59.02±6.30	96.36±4.98	98.48±2.62
H5	60.24±2.96	87.56±4.93	76.67±4.17	77.15±0.88	**91.66±0.60**	62.06±6.29	81.30±3.07
UNK	67.16±1.22	34.28±4.37	73.62±2.43	56.92±4.72	64.23±2.79	**79.72±3.10**	77.90±2.05
OS(%)	56.88±0.80	62.35±1.41	71.90±1.75	71.29±0.82	69.24±1.54	75.16±1.02	**79.26±1.58**
OS*(%)	54.83±0.97	67.97±1.99	71.55±1.78	74.17±0.71	70.24±1.57	74.25±1.40	**79.47±1.64**
HOS(%)	60.36±0.72	45.39±3.66	72.56±1.87	64.41±2.29	67.11±1.91	76.85±1.27	**78.71±1.73**
Kappa	44.06±1.18	41.75±1.62	57.67±0.71	51.16±1.25	59.42±1.36	58.66±2.34	**66.82±1.90**

TABLE III
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (BOTSWANA DATASET)

Class	SCLUDA	MLUDA	MSDA	OSLPP	SSOUDA	UADAL	CRLN
B1	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**	**100±0**
B2	95.29±1.40	97.15±1.63	96.13±2.63	98.01±0.72	**98.70±0.72**	97.08±0.61	97.16±1.36
B3	55.36±6.57	50.15±6.25	27.58±5.88	57.10±7.94	62.38±7.16	63.44±7.97	**63.70±3.27**
B4	**100±0**	99.75±0.48	**100±0**	**100±0**	98.21±2.13	**100±0**	99.25±1.50
B5	83.17±5.38	76.46±3.63	**99.28±0.52**	88.92±2.22	89.73±9.36	73.35±5.27	78.92±4.64
B6	54.73±8.25	62.57±7.93	**69.42±9.77**	48.46±6.82	44.75±8.05	50.62±6.44	60.03±4.86
UNK	52.71±1.94	45.04±2.52	28.90±7.71	52.53±7.98	55.75±2.31	65.63±4.3	**72.47±1.83**
OS(%)	77.31±0.58	75.87±1.48	74.42±2.24	77.96±0.84	78.76±1.22	78.59±0.75	**81.65±0.73**
OS*(%)	81.42±0.76	81.01±1.84	82.01±1.55	82.20±0.47	82.59±1.66	80.75±1.32	**83.18±0.89**
HOS(%)	63.97±1.34	57.84±1.96	42.32±8.43	64.10±4.48	66.57±0.95	72.32±2.24	**77.44±1.01**
Kappa	67.72±0.72	65.02±1.49	60.21±3.69	68.60±2.37	70.04±1.11	71.53±0.92	**76.12±0.89**

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (PAVIA DATASET)

Class	SCLUDA	MLUDA	MSDA	OSLPP	SSOUDA	UADAL	CRLN
P1	87.46±0.57	98.20±0.44	94.52±1.07	**98.32±1.62**	96.11±0.24	94.50±0.30	98.00±0.48
P2	55.96±2.03	41.17±5.35	55.46±5.36	**92.78±3.26**	45.32±2.44	52.39±8.90	61.07±1.31
P3	89.48±4.06	90.22±5.78	85.23±5.17	61.16±2.44	39.51±9.46	69.71±3.67	**92.33±1.06**
P4	81.80±1.36	75.59±6.77	81.93±8.32	79.14±6.48	**98.07±0.09**	60.49±0.27	90.41±1.91
P5	89.86±1.55	98.56±0.75	94.47±3.34	99.95±0.07	97.57±0.46	**100±0**	97.57±0.96
UNK	26.97±1.93	35.06±4.72	25.17±6.26	61.48±0.07	**65.96±0.67**	59.33±1.58	64.01±3.63
OS(%)	71.92±0.32	73.13±0.53	72.80±1.47	82.14±1.24	73.76±1.67	72.74±1.83	**83.88±0.35**
OS*(%)	80.92±0.31	80.75±0.41	82.32±2.83	86.27±1.47	75.31±2.08	75.42±2.52	**87.85±0.47**
HOS(%)	40.43±2.14	48.73±4.43	38.18±7.04	71.72±0.65	70.33±0.73	66.39±0.02	**74.01±2.29**
Kappa	31.33±0.93	33.19±1.96	30.26±1.15	58.10±0.23	49.74±1.19	47.23±2.45	**58.25±2.38**

The optimal parameters are not consistent between different datasets, but for simplicity, we set the parameters α, β_1, β_2, and τ of the three datasets consistent, that is, $\alpha = 1e^{-1}$, $\beta_1 = 0.5$, $\beta_2 = 0.5$, and $\tau = 0.07$. As can be seen from Tables II–IV, the proposed network can still achieve the best performance on the three datasets under the above parameter settings.

D. Comparative Experiment and Analysis

From Tables II–IV, we can see that:

1) Compared with unsupervised domain adaptation methods (SCLUDA, MLUDA, and MSDA), open set domain adaptation methods (OSLPP, SSOUDA, UADAL, and CRLN) generally achieve better classification performance, this is because: Unsupervised closed set domain adaptation methods have difficulty in handling unknown classes. Simply setting a threshold does not achieve remarkable classification performance, and inappropriate selection of a threshold can lead to excessively high or low accuracy for unknown classes, and the same is true for the accuracy of known classes. In addition, on some datasets, such as Houston dataset, the closed set method MSDA achieved good performance. This

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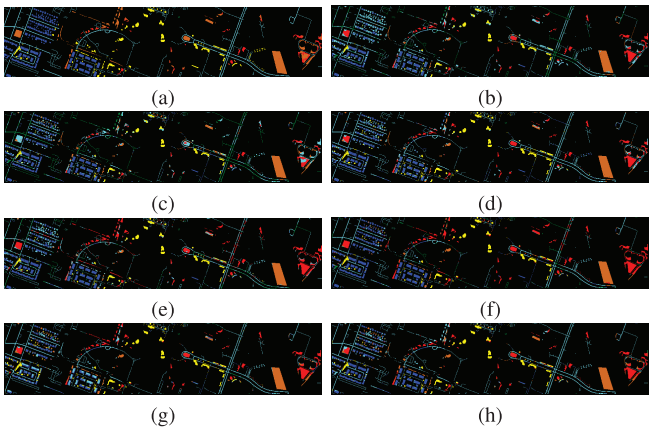


Fig. 6. Classification accuracy maps on Houston dataset. (a) Ground truth. (b) SCLUDA. (c) MLUDA. (d) MSDA. (e) OSLPP. (f) SSOUDA. (g) UADAL. (h) CRLN.

is because it combines adversarial training and self-distillation learning during the training process, fully utilizes the class-related knowledge in the target-domain data, and improves the separability of features, so even a simple threshold setting still has high accuracy.

2) Among the open set domain adaptation methods (OSLPP, SSOUDA, UADAL, and CRLN), OSLPP, SSOUDA, and UADAL have performed well on different datasets, maintaining an upper-middle level of accuracy for both known and unknown classes. In contrast, CRLN not only maintains a high accuracy for known classes but also achieves a high accuracy for unknown classes. This is because: on the one hand, CRLN can learn the discriminative feature representation of known and unknown classes, while mining the incorrect knowledge of the teacher network, and the student network then learns reflectively on this part of incorrect knowledge. On the other hand, CRLN achieves an effective source domain knowledge transfer effect by aligning the known classes in the target-domain output by the student network with the source domain and pushing the unknown classes away from the known classes.

E. Visual Analysis

1) Classification Accuracy Maps: In this section, we visualize the classification accuracy of CRLN and its comparison methods on the three datasets. As shown in Figs. 6–8, we annotate the ground truth and method names above or below the corresponding visualization results. It can be seen that CRLN has the best classification accuracy on the three datasets.

2) t-SNE Maps: In this section, we show the t-SNE plots of the original HSI and CRLN output features in Fig. 9. From these plots, we can observe that CRLN accurately captures the key class identification information in HSI. Its output features have high intraclass consistency and interclass separability.

F. Ablation Study

CRLN consists of three key components: EDM, FKEM, and CASM. We constructed the ablation models and conducted

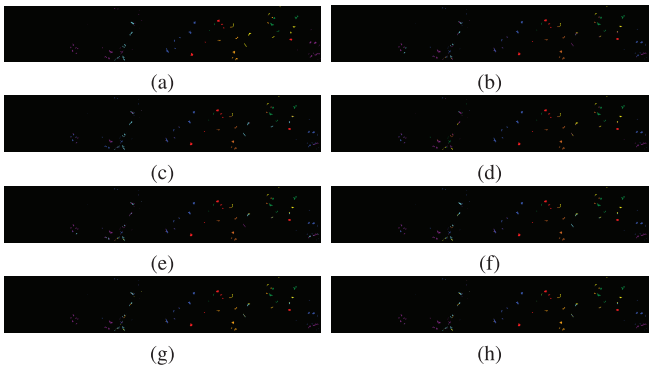


Fig. 7. Classification accuracy maps on Botswana dataset. (a) Ground truth. (b) SCLUDA. (c) MLUDA. (d) MSDA. (e) OSLPP. (f) SSOUDA. (g) UADAL. (h) CRLN.

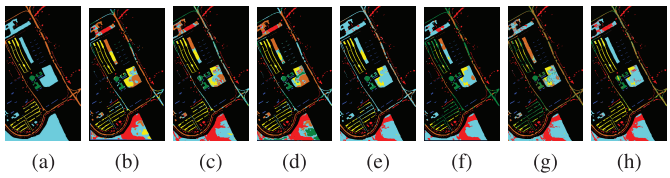


Fig. 8. Classification accuracy maps on Pavia dataset. (a) Ground truth. (b) SCLUDA. (c) MLUDA. (d) MSDA. (e) OSLPP. (f) SSOUDA. (g) UADAL. (h) CRLN.

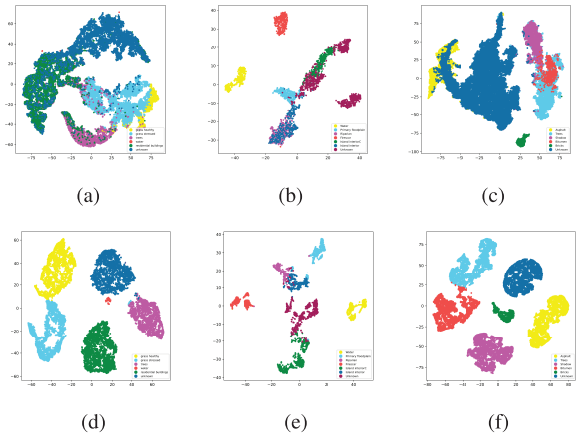


Fig. 9. T-SNE maps. (a) t-SNE map of original HSI on Houston 2018 dataset. (b) t-SNE map of original HSI on BOT6 dataset. (c) t-SNE map of original HSI on PaviaU dataset. (d) t-SNE map of output features learned by CRLN on Houston 2018 dataset. (e) t-SNE map of output features learned by CRLN on BOT6 dataset. (f) t-SNE map of output features learned by CRLN on PaviaU dataset.

ablation experiments to verify the effectiveness of each component, as shown in Tables V and VI. Next, we will introduce the construction principles of each ablation model.

CRLN-A: This model uses the teacher network output as the reference label for the student network output and updates the model using cross entropy loss, using only domain adversarial adaptation. After training, the target-domain HSI is directly classified.

CRLN-B: Based on CRLN-A, this model aligns known classes into subdomains and uses contrastive learning to bring the same subdomains closer and push the distances between

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different subdomains farther, thereby achieving adaptation between different domains.

CRLN-C: Based on CRLN-B, this model adds FKEM. The student network learns samples that the teacher network predicts incorrectly and optimizes them using a distance-based loss. Then, it completes the HSI classification task for the target domain.

CRLN-D: Based on CRLN-B, EDM is added to this model. Student network learns through the probability prediction of the known class and the unknown class given by the teacher network, and the similarity of the output of the teacher–student network is measured by calculating the KL divergence.

CRLN-E: This model is based on CRLN-A with FKEM and EDM added. The student network not only learns from the teaching process of the teacher network but also through the fallible knowledge of the teacher network found by FKEM and then performs reflective learning.

CRLN: Based on CRLN-E, this model adds CASM. After the teacher network guides the student network to learn and the student network learns reflectively based on the teacher network's fallible knowledge, it brings the same subdomains closer and pushes the distances between different subdomains farther, thus achieving the open set cross-domain HSI classification task.

From Table VI, we can see that:

1) Compared with CRLN-A, CRLN-B improves the classification accuracy by 0.74%, 1.19%, and 0.66% on the three datasets, respectively. This is because CRLN-B aligns the known classes of the source and target domains through subdomain alignment and uses contrastive learning to push away unknown classes, which can narrow the data distribution difference between the source and target domains and ensure the adaptability of the model to changes in domain distribution.
2) Compared with CRLN-B, CRLN-C improves HOS by 1.74%, 0.56%, and 0.52% on the three datasets, respectively. This is because CRLN-C can learn from the mistakes of the teacher network and optimize using distance-based loss in the prototype space, thus improving the classification accuracy of the model for all classes.
3) Compared with CRLN-B, CRLN-D improves HOS by 2.74%, 0.09%, and 0.85% on the three datasets, respectively. This is because CRLN-D constrains the student network to be similar to the teacher network that is good at classifying known classes, so that the student network can learn both the knowledge of known classes and the knowledge of unknown class classification, further improving the classification performance of the model in the open set cross-domain HSI classification task.
4) Compared with CRLN-A, CRLN-E improves HOS by 3.72%, 2.66%, and 2.60% on the three datasets, respectively. This is because CRLN-E can not only learn the knowledge correctly output by the teacher network but also extract heuristic knowledge from samples misclassified by the teacher network and optimize it using distance-based loss in the prototype space, further improving the classification accuracy of the model for known and unknown classes.
5) Compared with CRLN-E, CRLN improves HOS by 0.94%, 0.65%, and 0.24% on the three datasets, respectively. This is because CRLN, based on CRLN-E, aligns the known classes of the source domain and target domain finally classified by the student network into subdomains and uses contrastive learning to push the unknown classes away. This method can effectively narrow the data distribution difference between the source domain and the target domain, thereby improving the classification performance of the model in the open set cross-domain HSI classification task.

TABLE V
ABLATION MODELS

Component	CRLN-A	CRLN-B	CRLN-C	CRLN-D	CRLN-E	CRLN
CASM	✘	✔	✔	✔	✘	✔
FKEM	✘	✘	✔	✘	✔	✔
EDM	✘	✘	✘	✔	✔	✔

TABLE VI
IMPACT OF DIFFERENT COMPONENTS ON HOS (%)

Dataset	CRLN-A	CRLN-B	CRLN-C
HS13-HS18	74.05±0.49	74.79±1.20	76.53±0.65
BOT5-BOT6	74.13±0.39	75.32±1.00	75.88±0.80
PaviaC-PaviaU	71.17±0.79	71.83±0.66	72.35±1.13
Dataset	CRLN-D	CRLN-E	CRLN
HS13-HS18	77.53±0.52	77.77±0.51	**78.71±1.73**
BOT5-BOT6	75.41±0.32	76.79±1.20	**77.44±1.01**
PaviaC-PaviaU	72.68±1.68	73.77±0.70	**74.01±2.29**

TABLE VII
COMPUTATIONAL COST ON EACH DATASET WITH DIFFERENT METHODS

Dataset		SCLUDA	MLUDA	MSDA	SSOUDA	UADAL	CRLN
Houston	Time(s)	0.40	0.66	1.23	0.34	10.51	**0.32**
	FLOPs(M)	954.32	1967.45	965.13	954.95	979.78	**949.48**
	params(M)	1.75	2.10	**0.60**	1.76	2.15	1.67
BOT	Time(s)	0.12	0.22	0.21	**0.10**	1.61	0.15
	FLOPs(M)	2945.75	5950.34	**1360.26**	2945.38	2971.23	2940.12
	params(M)	1.98	2.33	**0.75**	1.99	2.37	1.90
Pavia	Time(s)	2.57	1.28	2.41	1.08	19.68	**0.78**
	FLOPs(M)	2525.69	162.54	966.21	2052.30	2542.43	**77.33**
	params(M)	1.87	2.23	**0.69**	1.88	1.79	1.77

G. Complexity Analysis

In this section, we add complexity analysis of different methods, including inference time, floating point operations (FLOPs), and number of network parameter. The results are shown in Table VII.

It can be seen from Table VII that compared with SCLUDA, MLUDA, MSDA, SSOUDA, and UADAL, CRLN has the least inference time and FLOPs on the Houston and Pavia datasets, and it ranks in the top three levels of inference time, FLOPs, and parameters on the three datasets. This is because CRLN uses a teacher–student network, the student network inherits the knowledge of the teacher network while maintaining a simple network structure. In summary, CRLN achieves the highest classification accuracy with a slight increase in computational cost.

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TABLE VIII
CLASSES AND THEIR NUMBER OF BOTSWANA DATASET (K = 1–3)

	Class	Name	BOT5 (Source) Number	BOT6 (Target) Number
	B1	Water	297	361
	B2	Primary floodplain	437	308
	B3	Riparian	448	303
	B4	Firescar	354	335
	B5	Island interior	337	370
	B6	Woodlands	357	324
K=1	Unk	Savanna	0	342
K=2	Unk	Savanna	0	342
		Short mopane	0	299
K=3	Unk	Savanna	0	342
		Short mopane	0	299
		Exposed soils	0	229

TABLE IX
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (BOTSWANA DATASET) (K = 1)

	SCLUDA	MLUDA	MSDA	SSOUDA	UADAL	CRLN
UNK	83.23±2.34	69.14±6.72	20.68±2.52	94.15±5.08	93.66±5.66	**97.37±2.72**
OS(%)	79.19±1.36	79.98±0.76	72.43±2.07	81.33±1.12	82.12±1.88	**85.61±1.42**
OS*(%)	78.52±1.46	81.78±2.01	81.05±2.03	79.20±0.62	80.20±1.45	**83.65±1.58**
HOS(%)	80.79±1.46	74.80±3.10	32.92±3.35	85.99±2.41	87.35±1.74	**89.97±1.51**
Kappa	76.81±1.63	77.67±0.81	68.55±2.44	79.31±1.28	80.07±2.17	**84.02±1.57**

TABLE X
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (BOTSWANA DATASET) (K = 2)

	SCLUDA	MLUDA	MSDA	SSOUDA	UADAL	CRLN
UNK	61.95±3.40	56.26±5.02	38.61±3.42	61.15±9.69	63.55±1.83	**64.37±2.12**
OS(%)	81.26±1.29	80.86±1.39	71.30±4.12	79.75±3.34	79.50±1.25	**82.74±1.24**
OS*(%)	84.48±1.49	84.07±2.18	76.04±3.92	82.85±3.47	82.16±1.19	**85.86±1.72**
HOS(%)	71.48±2.07	66.03±2.35	51.09±2.96	70.37±6.29	68.90±4.23	**73.34±0.45**
Kappa	75.81±1.52	74.15±1.02	62.08±3.30	74.16±3.74	74.24±1.52	**77.71±1.04**

TABLE XI
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS (BOTSWANA DATASET) (K = 3)

	SCLUDA	MLUDA	MSDA	SSOUDA	UADAL	CRLN
UNK	52.71±1.94	45.04±2.52	28.90±7.71	55.75±2.31	65.63±4.30	**72.47±1.83**
OS(%)	77.31±0.58	75.87±1.48	74.42±2.24	78.76±1.22	78.59±0.75	**81.65±0.73**
OS*(%)	81.42±0.76	81.01±1.84	82.01±1.55	82.59±1.66	80.75±1.32	**83.18±0.89**
HOS(%)	63.97±1.34	57.84±1.96	42.32±8.43	66.57±0.95	72.32±2.24	**77.44±1.01**
Kappa	67.72±0.72	65.02±1.49	60.21±3.69	70.04±1.11	71.53±0.92	**76.12±0.89**

H. Experiments With Different Numbers of Unknown Classes

In this section, we discuss the performance of the CRLN with different numbers of unknown classes. We conducted an experiment using the BOT dataset as an example. In the experiment, we set the number of unknown classes K = 1–3, respectively. The specific experimental settings and the classification performance are shown in Tables VIII–XI.

From the results, we can see that:

1) Under different K values, the HOS of SCLUDA, MLUDA, MSDA, SSOUDA, and UADAL are all lower than CRLN. Among them, SCLUDA, MLUDA, and MSDA are closed-set domain adaptation methods. Although they have achieved good results in classifying known classes, they cannot effectively identify and distinguish unknown classes. SSOUDA and UADAL cannot effectively extract the spatial information of HSIs, so they are not as stable and efficient as CRLN overall.
2) From the experimental results of K = 1 to K = 3, it can be seen that the CRLN achieves the highest HOS under different numbers of unknown classes. This shows that CRLN has significant advantages in the open set cross-domain HSI classification task, can effectively identify unknown classes, and can maintain remarkable performance even when there are multiple unknown classes.

In general, the above experimental results can verify that the CRLN can still achieve satisfactory accuracy when there are multiple unknown classes.

V. CONCLUSION

The CRLN framework for open set cross-domain HSI classification is proposed to address the problems of identifying unknown class and distribution differences. In CRLN, a teacher–student network is introduced into the open set domain adaptation task, and the teacher network trained only on the source domain is enabled to explicitly distinguish unknown class by measuring uncertainty, and the student network learns heuristic knowledge from the teacher network's error experience, which enhances the classification ability of the model in the target domain. At the same time, the known class samples of the target domain are aligned with the source data, and the unknown class samples are pushed away by contrastive learning. Experiments on three datasets show that our method achieves better results than the existing methods, proving the effectiveness of our method. Future research could consider generating pseudo-unknown samples and explore additional strategies tailored to the characteristics of unlabeled HSI data in the target domain to further improve the performance of cross-scene HSI classification.

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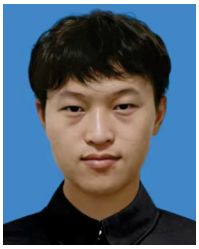
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